AI 101 for Mechanical Engineers · Module 15

Uncertainty and changing conditions

Lesson 15.1

What you will learn

> Distinguish aleatoric from epistemic uncertainty, explain why confidence may be unreliable outside evaluated conditions, relate distribution shift and drift, and decide when to monitor, defer or gather evidence.

Lesson overview

Module 15 · Lesson 1

Uncertainty and changing conditions

AI 101 for Mechanical Engineers

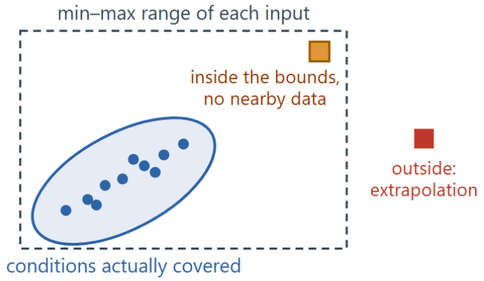
1 Two sources of uncertainty

Aleatoric
Variation in the observations, such as sensor noise in repeated readings of a steady temperature. Repeated measurements describe it but cannot remove it.

Epistemic
Limited knowledge, such as sparse data in one operating regime. More informative data can reduce it, but cannot fix a flawed model.

Calibration checks whether stated probabilities match observed frequencies for a group of cases. It is not a promise about one prediction.

2 Coverage is more than input bounds



Inside the range is not a reliability guarantee. Interpolation still needs representative evidence and validation.

High model confidence does not justify extrapolation.

3 Distribution shift and drift

Distribution shift: the data distribution differs between the conditions compared, e.g. a model developed at one plant type and used at another.

Drift: change over time. It can be



abrupt – a sensor replaced by one calibrated differently

gradual – a sensor slowly wearing

recurring – an earlier pattern returns

Concept drift: the relationship between inputs and target changes. The terms overlap: drift creates a shift between earlier and later data.

A random seed cannot correct shift or drift.

Key takeaway

Know where the evidence ends, and watch for change.
Set monitoring, deferral and revalidation rules before deployment.

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Uncertainty has two different sources

Aleatoric

- Variation, e.g. sensor measurement noise [79]
- Repeats may characterize it, not remove it

Epistemic

- Comes from limited knowledge
- e.g. sparse measurements, incomplete examples, possibly unrepresentative assumptions
- More informative observations can reduce some

Observations cannot repair inadequate model structure; a flawed model's confident answer may still be wrong.

Interpolation is better supported than extrapolation

Interpolation

- Within conditions covered by relevant observations

Extrapolation

- Beyond the covered operating domain
- Behavior is less well supported

Inside every input range is not enough

- A point can lie between each input’s minimum and maximum
- Yet input combinations, plant configurations or regimes may be missing
- Define the domain by conditions relevant to the intended use
- Evaluate performance on evidence from that domain

Calibration is a group-level property

- Checks stated probabilities against observed frequencies [12]
- Calibrated model: many comparable cases at about 80% are about 80% correct
- Not a promise about, or explanation of, one prediction
- Assess on independent data from intended operating conditions

> High confidence is not proof of calibration, accuracy or permission to extrapolate.

Distribution shift and drift overlap

Distribution shift

- Data differ between compared conditions [21]
- e.g. one plant type for development, another for deployment

Drift

- Change in data or relationships over time
- Concept drift: input–target relationship changes [59, Sec. 2.1]

> Drift over time produces a shift between earlier and later data. A random seed corrects neither.

Drift can take different forms

Form	Example
Abrupt	Sensor replaced by one calibrated differently
Gradual or incremental	A sensor slowly wearing
Recurring	An earlier pattern returns

Forms of drift as described in [59, Sec. 2.2].

Monitor, then act on rules set in advance

1. Compare inputs, missingness, outputs and alerts with development conditions [59]
2. Input changes are warnings, not proof by themselves of degraded accuracy
3. Check outcomes when trustworthy labels become available
4. Investigate, obtain validation evidence, or recalibrate where justified
5. Retrain only with appropriate data; defer outside validated domain

CHECK YOUR UNDERSTANDING

Can high model confidence justify using a model outside its validated operating domain?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Can high model confidence justify using a model outside its validated operating domain?

> No. Confidence is not proof of calibration or accuracy, and calibration is checked only for evaluated conditions. Outside it, defer or gather new validation evidence.

CHECK YOUR UNDERSTANDING

A pump model developed at one plant is moved to a different type of plant. Is this drift or distribution shift?

Answer before moving on.

CHECK YOUR UNDERSTANDING

A pump model developed at one plant is moved to a different type of plant. Is this drift or distribution shift?

> Distribution shift: deployment data come from a different population from development data. Drift is change over time, e.g. sensor ageing.

Key takeaways

- Aleatoric: variation in observations; epistemic: limited knowledge
- Define the operating domain by relevant conditions, not only input limits
- Calibration is group-level; high confidence is not proof of accuracy
- Shift: data distributions differ; drift: change over time
- Set monitoring, deferral and revalidation rules before deployment

References

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[59] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, and A. Bouchachia, “A survey on concept drift adaptation,” ACM Comput. Surv., vol. 46, no. 4, pp. 1–37, 2014. doi: 10.1145/2523813

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Numbers match the course reference list. The written lesson is the authoritative text.